

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Course 6 – Computer Science & Engineering (6-3)

Four-Year Academic Plan with Core Textbooks

Bachelor of Science Degree Program

Overview

MIT's Course 6-3 (Computer Science and Engineering) is one of the most rigorous and prestigious undergraduate CS programs in the world. This plan follows the standard 6-3 curriculum, combining foundational mathematics, software engineering, systems, algorithms, and AI/ML electives into a coherent four-year sequence.

Credit System: MIT uses units (not credits). Most classes carry 12 units (roughly 12 hrs/week effort). A typical semester load is 48–57 units (4–5 classes). Graduation requires 360 units total, including HASS (Humanities, Arts & Social Sciences) and CI (Communication Intensive) requirements.

Year 1 – Foundations

The first year establishes the mathematical and programming foundations that all later coursework builds upon. Students take calculus, physics (GIR requirements), and introductory programming.

Fall Semester

Course	Course Name	Units	Textbook(s)
6.1010	Fundamentals of Programming	12	How to Think Like a Computer Scientist (Downey)
18.01	Single Variable Calculus	12	Calculus (Stewart)
8.01	Physics I – Classical Mechanics	12	University Physics (Young & Freedman)
CI-HW	Communication Intensive (HASS)	9	Varies by section

Spring Semester

Course	Course Name	Units	Textbook(s)
6.1020	Elements of Software Construction	12	Software Abstractions (Jackson)
18.02	Multivariable Calculus	12	Calculus (Stewart), Vol. 3
8.02	Physics II – Electricity & Magnetism	12	University Physics (Young & Freedman)
HASS	HASS Elective	9	Varies by course

Year 2 – Core Computer Science

Year 2 dives into the theoretical and systems core of computer science: discrete mathematics, algorithms, computer architecture, linear algebra, and probability. These are prerequisites for nearly all upper-level 6.x courses.

Fall Semester

Course	Course Name	Units	Textbook(s)
6.1200	Mathematics for Computer Science	12	Mathematics for Computer Science (Lehman, Leighton & Meyer)
6.1210	Introduction to Algorithms	12	Introduction to Algorithms (CLRS)
6.1910	Computation Structures	12	Digital Design & Computer Architecture (Harris & Harris)
18.06	Linear Algebra	12	Introduction to Linear Algebra (Strang)

Spring Semester

Course	Course Name	Units	Textbook(s)
6.1800	Computer Systems Engineering	12	Computer Systems: A Programmer's Perspective (Bryant & O'Hallaron)
6.3700	Introduction to Probability	12	Probability and Statistics for Engineers (DeGroot & Schervish)
6.1060	Software Performance Engineering	12	Computer Architecture (Hennessy & Patterson)
HASS	HASS Elective	9	Varies by course

Year 3 – Advanced Topics

Year 3 introduces specialization tracks including artificial intelligence, database systems, security, and parallel computing. Students begin to explore research areas and satisfy their CI-M (Communication Intensive in the Major) requirement.

Fall Semester

Course	Course Name	Units	Textbook(s)
6.1820	Mobile & Sensor Computing	12	Wireless Communications (Goldsmith)
6.3800	Introduction to Inference	12	Pattern Recognition & Machine Learning (Bishop)
6.4100	Artificial Intelligence	12	Artificial Intelligence: A Modern Approach (Russell & Norvig)
6.5830	Database Systems	12	Database System Concepts (Silberschatz, Korth & Sudarshan)

Spring Semester

Course	Course Name	Units	Textbook(s)
6.5081	Multicore Programming	12	The Art of Multiprocessor Programming (Herlihy & Shavit)
6.5610	Applied Cryptography & Security	12	Introduction to Modern Cryptography (Katz & Lindell)
6.4110	Representation, Inference & Reasoning in AI	12	Probabilistic Graphical Models (Koller & Friedman)
CI-M	Communication Intensive (Major)	12	Varies by course

Year 4 – Capstone & Electives

The final year centers on advanced topics, independent research (6.UAT/6.UAP), and deep technical electives. Students complete a supervised research project, often leading to a thesis or conference paper. HASS and free elective requirements are typically completed here.

Fall Semester

Course	Course Name	Units	Textbook(s)
6.S898	Graduate Deep Learning (or undergrad equiv.)	12	Deep Learning (Goodfellow, Bengio & Courville)
6.5900	Computer Architecture	12	Computer Architecture: A Quantitative Approach (Hennessy & Patterson)
6.UAT	Undergrad Advanced Project (Thesis I)	6	N/A – Research project
HASS	HASS Elective	9	Varies by course

Spring Semester

Course	Course Name	Units	Textbook(s)
6.UAP	Undergrad Advanced Project (Thesis II)	12	N/A – Research project
6.5940	Hardware Architecture for Deep Learning	12	Efficient Processing of Deep Neural Networks (Sze et al.)
Elective	Technical Elective (e.g. 6.5660 Compilers)	12	Engineering a Compiler (Cooper & Torczon)
HASS	HASS Elective	9	Varies by course

Essential Textbook Library

The following textbooks are considered core references across the 6-3 curriculum. Students are strongly encouraged to own or have permanent access to these titles.

Textbook	Authors	Used In
Introduction to Algorithms (CLRS), 4th Ed.	Cormen, Leiserson, Rivest & Stein	6.1210
Mathematics for Computer Science	Lehman, Leighton & Meyer	6.1200
Computer Systems: A Programmer's Perspective	Bryant & O'Hallaron	6.1800
Artificial Intelligence: A Modern Approach	Russell & Norvig	6.4100
Pattern Recognition & Machine Learning	Bishop	6.3800
Deep Learning	Goodfellow, Bengio & Courville	6.S898
Database System Concepts	Silberschatz, Korth & Sudarshan	6.5830
Introduction to Linear Algebra	Gilbert Strang	18.06
Introduction to Modern Cryptography	Katz & Lindell	6.5610
Computer Architecture: A Quantitative Approach	Hennessy & Patterson	6.5900
The Art of Multiprocessor Programming	Herlihy & Shavit	6.5081
Probabilistic Graphical Models	Koller & Friedman	6.4110

Notes & Advisement

1. Course numbers changed in 2023 – MIT renumbered most 6.xxx courses. This plan uses the new numbering scheme. Verify current offerings at student.mit.edu.
2. Elective flexibility – Course 6-3 allows considerable elective freedom in years 3–4. Students may substitute technical electives in robotics (6.4210), computer vision (6.8300), NLP (6.8611), or other tracks based on interest.
3. HASS requirement – 8 HASS subjects required total, including 1 CI-H and at least 1 concentration. These are distributed across all four years.
4. Units – 360 units required for graduation. This plan totals approximately 360–384 units depending on elective choices.
5. MIT OpenCourseWare (ocw.mit.edu) provides free lecture notes, assignments, and some textbooks for most of these courses.

Prepared as a reference guide. Always confirm current requirements with your MIT academic advisor and the official MIT Course Catalog.